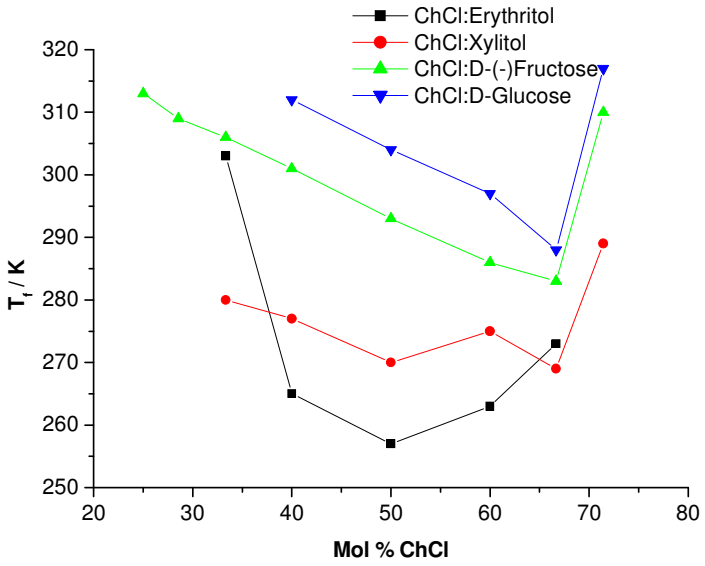


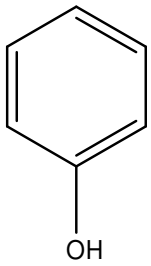

***Figure 3.45**: Phase diagram of ChCl:Polyol mixtures*

From the data shown in Figure 3.45, there are some interesting trends with regards to the number of OH groups there are in the polyol. It can be seen that DES's can be formed in molar ratios where ChCl is in molar excess. The results for the 4 OH containing species erythritol (14), show that the eutectic point occurs at a ratio of 1:1 ChCl:polyol. As the number of OH groups increases to 5, the eutectic point shifts from a ratio of 1:1 to a ratio of 2:1 ChCl:polyol as seen in the D-(-) fructose (16) and D-glucose (17) cases. What is perhaps most interesting about the data shown in Figure 3.44 is that the data for xylitol (15) shows two inflection points suggesting two eutectic points. These point occur at 1:1 which is the same as the erythritol system and at 2:1, the same seen in the D-(-) fructose (16) and D-glucose (17) systems. The erythritol system is thought to form the eutectic at a ratio of 1:1 due to the ChCl being able to hold 4 OH interactions, analogous to the eutectic of diols being at a ratio of 1:2 (2 x 2 OH = 4 OH). Since xylitol, D-(-) fructose and D-glucose all contain 5 OH groups more than one ChCl molecule is thought to interact. Because both D-(-) fructose and D-glucose are in cyclic form the angle of interaction between the Cl anion and OH groups is more favourable to allow 2 ChCl molecules to form hydrogen bonds. This would explain the larger depression in freezing point than in the case of xylitol. Because of the straight chain nature of xylitol it is possible to allow 2 Cl anions to form hydrogen bonds with

OH groups however the repulsion between ChCl molecules is likely to reduce this interaction.

## 3.6 Mono Functional Alcohol Deep Eutectic Solvents

This chapter has gone into a lot of detail about the use of alcohols with two or more –OH groups to form DES's but it is possible to form a DES from choline chloride and phenol which contains only one –OH group.




(18)

From the viscosity and conductivity data shown in Figures 3.45 and 3.46 respectively, the eutectic point appears to be at a ratio of 4:1 (Phenol:ChCl). The surprising result is perhaps the viscosity of these systems, where the eutectic point exhibits a viscosity of 30 cP at 2 ºC. Due to the extremely low freezing point of some of these mixtures near the eutectic point (< -40 ºC), a phase diagram was not constructed due to the large degree of inaccuracy of measuring the freezing points at such low temperatures.

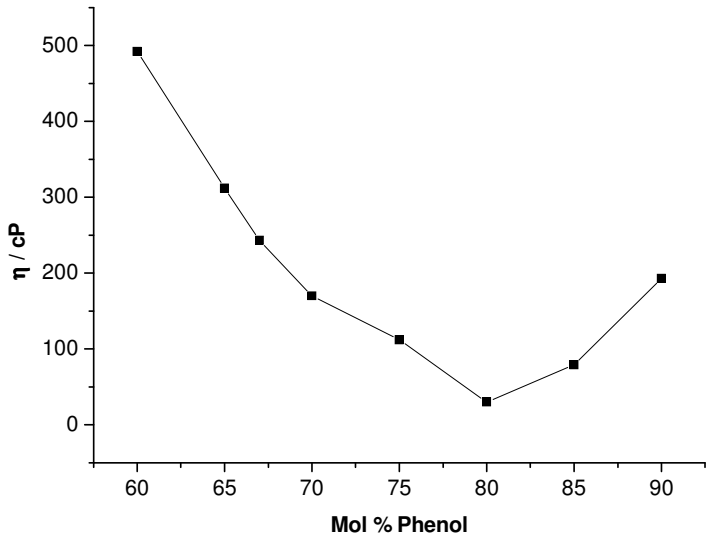



**Figure 3.45**: Viscosity of ChCl:Phenol mixtures as a function of composition at 275K

The viscosity of this system is amongst one of the least viscous DES's formed from two solid components. Currently it is thought that it offers the lowest viscosity of any ChCl/hydrogen bonding compound mixture. This is attributed to the electron-withdrawing effects of the delocalised π-electron system in the aromatic ring. Through the hydrogen bond between the OH and Cl formed, the aromatic group increases the amount of delocalisation of Cl anion from the choline cation, hence an increase in entropy of the system, which results in the lower viscosity.

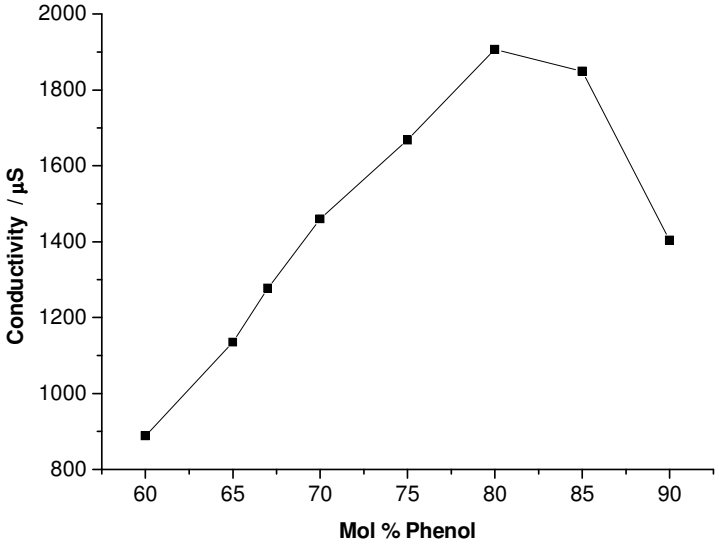



**Figure 3.46:** Conductivity of ChCl:Phenol mixtures as a function of composition at 275K

The conductivity, as shown in Figure 3.45, of this system reaches a maximum at a ratio of 1:4 (ChCl:Phenol), the same ratio at which the viscosity is at its lowest. From this it can be inferred that this is the eutectic point of this mixture.

## 3.7 Conclusions

This chapter has characterised the conductivity and fluidity of numerous ChCl/ alcohol based systems. A variety of alcohols form DESs but it was found that only those where the freezing point of the pure alcohol is close to or above ambient temperature do stable liquids form. A wide variety of polyols form stable liquids, including sugars. This offers the potential to create benign, even edible liquids from naturally occurring salts and sugars.

It has been shown that mixtures of quaternary ammonium salts with glycols form homogeneous fluids with the properties of ionic liquids when the mole fraction of the salt exceeds 0.2. At higher salt concentrations the dominant mode of charge transport is via the mobility of holes and the mixture has the viscosity and conductivity characteristic of an ionic liquid and below this concentration the properties can be described by the theories related to concentrated ionic solutions.

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# CHAPTER 4

# Investigation of Intramolecular Interactions in Deep Eutectic Solvents Using Solvatochromic Probes

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**4.1 Introduction**

*4.1.1 Solvatochromism*

*4.1.2 Kamlet-Taft Parameter, Polarisablility/Dipolarity π**

*4.1.3 Hydrogen Bonding - Hydrogen Bond Donor and Hydrogen Bond Acceptor Parameters α and β*

*4.1.4 $E_T(30)$ Betaine Dye Scale*

*4.1.5 Polarity of Ionic Liquids*

**4.2 Results and Discussion**

*4.2.1 $E_T(30)$ Betaine Dye Scale Results*

*4.2.2 Polarisability and Hydrogen Bond Acceptor/Donor Parameters π*, α and β*

*4.2.3 Effect of Choline Chloride Concentration*

**4.3 Conclusions**

**4.4 References**

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## 4.1 Introduction

Solvent polarity has in the past been measured by simple electrostatic models for intermolecular solute–solvent interactions.[1] By assuming the solvents to be a non-structured homogeneous continuum, dielectric constants, $\varepsilon_r$, dipole moments $\mu$, and refractive indices $n_D$, have been used as macroscopic summations of solvent polarity.[1] However, there is a fundamental flaw in these measurements. As solute–solvent interactions take place on a molecular–microscopic level, there are individual, mutually interacting solvent molecules surrounding the ions or molecules of the solute. This leads to solvation shells of a varying degrees of proximity to the solute molecule, commonly referred to as the cybotactic region.[2-9] Other specific interactions such as hydrogen-bond donating/accepting, electron pair donating/accepting or solvaophobic effects add to the poor correlation of solvent effects using the macroscopic solvent parameters.[1]

Since the 1970's there has been a considerable amount of research carried out to investigate the cybotactic region of the solute molecule.[2-9] Measurements of solvatochromatic shifts have been used to determine solute-solvent interactions such as the polarisability/dipolarity parameter, $\pi^*$, of the solvent as well as giving information about hydrogen bond donor (HBD), $\alpha$ and/or acceptor (HBA), $\beta$ abilities of the solvent.[9]

To understand the behaviour of a solvent involved in a reaction, it is important to understand the solute-solvent interactions of the system. Since this project is in the area of Deep Eutectic Solvents, this chapter investigates the polarisability and hydrogen bond donor/acceptor properties of several Deep Eutectic Solvents.

### 4.1.1 Solvatochromism

The ability of a chemical substance to change colour in a variety of solvent polarities is termed solvatochromism.[10] The change in colour is due to changes in the electronic absorption and emission spectra of the chemical substance. The displacement of the electronic spectrum for a specific solute in different solvents is known as a solvatochromic shift. There are two types of shift that can be observed in solvatochromism. A positive shift with increasing solvent polarity is termed bathochromic shift, or red shift, and a negative shift with increasing solvent polarity is

termed hypsochromic shift, or blue shift. The solvatochromic shift depicts the relationship of the absorption/emission spectra and the polarity of the solvent and gives information on the electronic excited states of the molecule. These electronic states each have a specific energy, denoted *E*. In quantum mechanics the Schrödinger equation relates this energy to the wavefunction, $\Psi$, and the Hamiltonian operator, *H*, by [11]

$$H\Psi = E\Psi \qquad \textbf{(4.1)}$$

The initial relaxed electronic states correspond to the most stable nuclear configurations of a molecule in an isolated environment. It is proposed by the Franck-Condon principle that the solvation pattern of both the initial relaxed and the first excited state, also known as the Franck-Condon state, are the same. This is because the time required to electronically excite the molecule (~$10^{-15}$ s), is much less than the time required for the molecule to carry out vibrational or rotational transitions (~$10^{-12}$ -$10^{-10}$ s).[12] This means that the nuclei of the absorbing molecule and the solvation sphere do not exhibit much change during the electronic transition. Reorientation of the solvation sphere will only occur if the lifetime of the excited state is large enough.

Figure 4.1 shows the processes involved in absorption/emission, $S_0 + h\upsilon \Leftrightarrow S_1$. In the process of absorption/emission the initial ($S_0$) and final ($S_1$) states are linked by a radiative transition. In Figure 4.1 it can be seen that the environment in which the molecule being excited is in, the energy levels in which ($S_0$) and ($S_1$) are affected. The first environment shown in Figure 4.1 is the absorption ($S_0 \rightarrow S_1$) of energy under gas phase conditions ($h\upsilon_{vac}$) and the second is that of the absorbing molecule surrounded by a solvent ($h\upsilon_{sol}$). The energy that stabilises these electronic states is known as the solvation energy, $E_s$. The polarity of the solvent is the determining factor in the solvation energy and it is this difference between the solvation energies of the initial and final state ($E^0_s$ and $E^1_s$ respectively) that causes the solvatochromic shift.

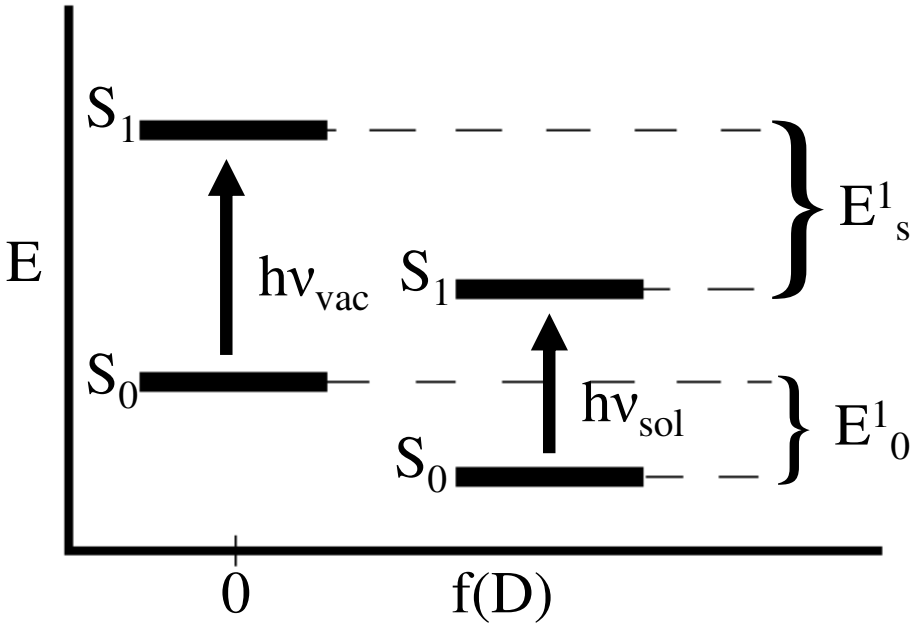



***Figure 4.1*** *Solvatochromic shift energy diagram for the case of a vacuum and a solvent f(D) is the Onsager solvent polarity function 2(ε -1)/(2ε + 1)*

A bathochromic shift results from the greater stability of the $S_1$ state by solvation, over the $S_0$ state. For a hypsochromic shift the $S_0$ state is stabilised more strongly than the $S_1$ state by solvation. By observing the solvatochromic shifts in various solvents, it is possible to describe the relative energies of the electronic states of a molecule. The data obtained from the shifts, can be used to determine a number of the physical properties of the molecule such as the dipole moment and polarisability, π*. In addition, information about specific interactions such as hydrogen bonding, α/β, can be obtained.

For a particular probe molecule to be considered useful for measuring these properties of the solvent, several requirements described by Reichardt must be met;[1]

1) The compound should be available by either easy synthesis or commercially

2) It should be in a crystalline form that has definite chemical structure, is easy to handle and is stable in storage, and more importantly stable in solution.

3) It should possess sufficient solubility in both non-polar and polar solvents.

4) There should be no chemical reaction with the solvent.

5) The extent to which the longest wavelength UV/vis/near-IR absorption band is shifted, either bathochromically or hypsochromically, with increasing solvent polarity, should be as large as possible.

### 4.1.2 Kamlet-Taft Parameter, Polarisablility/Dipolarity π*

The origins of the π* scale stems from a combination of different methods previously used to describe the polarity of a solvent. In 1948, Winstein introduced the *Y* scale.[13] This was the first empirical parameter used to describe the 'solvent ionising power'. By 1958 the first spectroscopic solvent polarity scale was developed by Kosower, known as the *Z* scale.[14] This scale was compiled by using the solvent-sensitive intermolecular charge-transfer (CT) absorption of 1-ethyl-4-(methoxycarbonyl)pyridinium iodide. There have since been many solvent polarity scales based on ultraviolet (UV)-visible or near infra-red (IR) spectroscopy developed.

The π* scale was developed by measuring the solvent induced shift of several nitroaromatic indicators and is one of the most popular scales employed to date.[15,16] The solvents were placed in order of longest to shortest wavelength seen in the $\pi \rightarrow \pi^*$ absorption band. The electronic transition is a result of intramolecular charge transfer from the electron donating group to the electron accepting group through the aromatic ring of the dye molecule. The π* scale currently has been normalised to give the range of 0.00 to 1.00 referring to cyclohexane and dimethyl sulfoxide respectively. These solvents were chosen because they are completely non-hydrogen bond donating. To prevent the inclusion of specific dye/solvent effects and spectral anomalies, more than one dye was used and the solvatochromic shifts averaged. The value obtained from the π* data arises from a measurement of dispersive, inductive and electrostatic dye/solvent interactions. This data therefore corresponds to a mix of both the dipolarity and the polarizability of the solvent.

The π* scale is dependant upon a linear solvation energy relationship (LSER) combining a number of possible solvent effects on a solute.[17,18] This relationship has the general form

$$XYZ = XYZ_0 + a\alpha + b\beta + SPPE \qquad \textbf{(4.2)}$$

where *XYZ* and $XYZ_0$ can represent a particular solvent-dependent property of the system for a given solvent and in a standard reference solvent respectively. These solvent-dependent properties may be reaction rates, equilibrium constants or a position/intensity of spectral absorption. The $\alpha$ term is a measure of the hydrogen bond donor (HBD)

interaction strength towards a solute. The $\beta$ term is a measure of the hydrogen bond acceptor (HBA) strength of the solvent.[15-20] The *a* and *b* parameters are the respective hydrogen bonding constants associated with the solute. The *SPPE* term refers to the solvent polarity/polarisability effect. This term can be broken down further and expressed by its constituent parts as

$$XYZ = XYZ_0 + yY + pP + eE + bB \quad \textbf{(4.3)}$$

where the *Y* term represents one or the other solvent "polarity functions" and the *P* term refers to the "polarizability function". The terms aα and bβ have been replaced by *eE* and *bB* which refer to electrophilicity and nucleophilicity respectively.

By using the LSER in equation 4.2, Kamlet and Taft *et al*. were able to correlate the characteristic values of the HBD, HBA and SPPE parameters for various solvents from the shift in peak position of a UV-visible absorption maximum of an indicator solute. They showed that the SPPE term could be simplified and expressed as a single parameter, π*, which is a measurement of the ability of the solvent to stabilize a neighbouring charge or a dipole by dielectric interactions. The term π* refers to the solvatochromic effects on p → π* and π → π* electronic spectral transitions.

With the SPPE term substituted by π* the equation becomes

$$XYZ = XYZ_0 + s(\pi^* + \partial\delta) + a\alpha + b\beta + h\delta_H \quad \textbf{(4.4)}$$

The *s* term is the susceptibility of *XYZ* to changing *SPP* and is dependant upon the solute. The term $\delta$ is a polarisability correction term that is given different values depending upon the nature of the solvent. For non-halogenated aliphatics, the value is equal to 0.0, for halogenated aliphatics the value is 0.5 and for aromatic solvents it is 1.0. When the electronic spectrum is bathochromically shifted, the coefficient of the $\delta$ term, $\partial$, is equal to zero and, therefore, $\delta$ can be neglected.

Equation 4.4 can be further simplified to equation 4.5 if the correct conditions are met. For equation 4.5 to work accurately α = β = 0 must be true. If the solvent is neither hydrogen bond acceptors nor hydrogen donators then this is true. If the solvent is a hydrogen bond acceptor (β ≠ 0) but is not a donator (α = 0) then 4.5 is true if the indicator solute is non-hydrogen bond donating (b = 0).

$$\upsilon_{max} = \upsilon_0 + s\pi^* \quad \textbf{(4.5)}$$

A selection of organic solvents with the respective π* values are shown in Table 4.1

***Table 4.1**: π* values for some selected organic solvents at room temperature and 1 bar pressure in ascending order.*[16]

| **Solvent** | **Solvent Type** | **π*** |
|---|---|---|
| Cyclohexane | **NHB** | 0.00 |
| Triethylamine | **HBA** | 0.14 |
| Carbon Tetrachloride | **NHB** | 0.28 |
| Ethanol | **HBA-D** | 0.54 |
| Ethyl Acetate | **HBA** | 0.55 |
| Tetrahydrofuran | **HBA** | 0.58 |
| Benzene | **Ar-NHB-HBA** | 0.59 |
| Methanol | **HBA-D** | 0.60 |
| Acetic Acid | **HBA-D** | 0.66 |
| Acetone | **HBA** | 0.71 |
| Acetonitrile | **HBA-HBD** | 0.75 |
| Dichloromethane | **NHB** | 0.80 |
| Pyridine | **Ar-HBA** | 0.87 |
| Dimethylformamide | **HBA** | 0.88 |
| Dimethyl Sulfoxide | **HBA** | 1.00 |
| Water | **HBA-D** | 1.09 |

**NHB** = non-hydrogen bonding solvent; **HBA** = hydrogen bond acceptor; **HBD** = hydrogen bond donor; **HBA-D** = amphiprotic hydrogen bond acceptor-donor; **Ar** = aromatic solvent

### 4.1.3 Hydrogen Bonding - Hydrogen Bond Donor and Hydrogen Bond Acceptor Parameters α and β

Other parameters that can be accounted for in the Kamlet-Taft method are the hydrogen bonding components of a given system α and β. These describe the hydrogen bond donating and accepting abilities respectively.

A donor-acceptor interaction that specifically involves a hydrogen atom is referred

to as a hydrogen bond. A hydrogen bond between two species can be either intermolecular or intramolecular. The common representation of a hydrogen bond is a broken line and can be seen in Figure 4.2. [21]

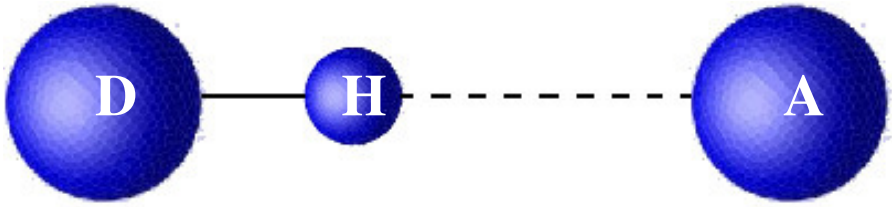



***Figure 4.2*** *Representation of a hydrogen bond between a donor and an acceptor*

A hydrogen bond can be formed between small and highly electronegative atoms, such as N, O or F. A species is capable of being a hydrogen bond donor when the electronegativity of the species, for example A in Figure 4.2, relative to H in an covalent bond is such that the withdrawal of electrons leaves the proton partially unshielded and the bond becomes capable of donating the proton. In order to interact with this donor bond, the acceptor species, for example B in Figure 4.2, must possess either a lone pair of electrons or polarisable π electrons. The strength of a hydrogen bond is found to be similar in strength to a van der Waals force, approximately 20 kJ $mol^{-1}$. The main difference between the two types of interaction is that a hydrogen bond is directional and this can give rise to discrete recognisable units which consist of two or more single molecules, whereas van der Waals forces are non directional. The hydrogen bonds found in compounds containing the groups -OH, $-NH_2$ or –NH- are an explanation for, among other effects, the higher boiling points than the isomeric molecules with no hydrogen directly attached to the oxygen or nitrogen atom. These stronger intermolecular forces result in larger lattice energies and hence require more energy to break the molecules apart.

Kamlet and Taft demonstrated that it is possible to measure the strength of a hydrogen bond by the UV-visible spectral data.[17,18] Similar to the method for calculating π*, a solvatochromic comparison method was used. From this method the α-scale of HBD acidities and β-scale of solvent HBA basicities were successfully constructed. By modifying equation 4.5 these hydrogen bonding interactions can be accounted for, as shown in equation 4.6

$$\upsilon_{max} = \upsilon_0 + s\pi^* + a\alpha + b\beta \qquad \textbf{(4.6)}$$

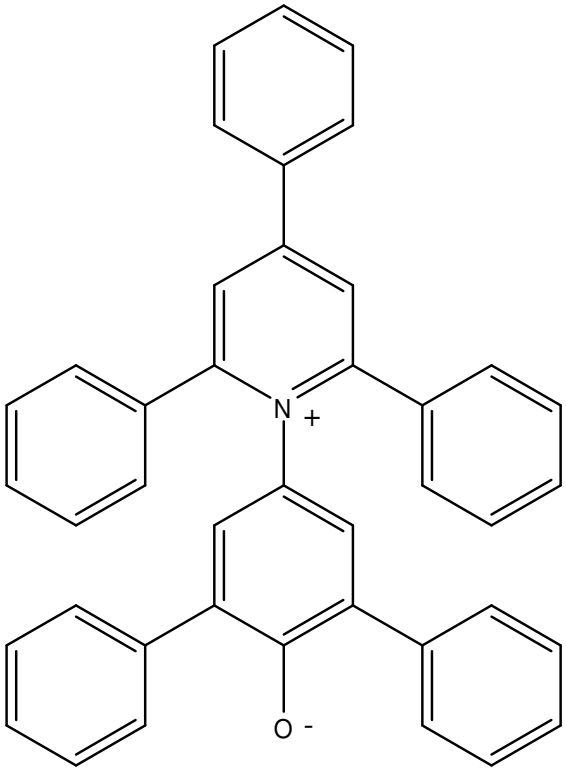
The solvatochromic comparison method can be used to determine α values by comparison of solvent-induced shifts of the longest wavelength $\pi \rightarrow \pi^*$ absorption band of two similar probe molecules. Only one of these must be able to act as a HBA towards HBD solvents. The corresponding β values may be determined by an analogous method, where one of the probe molecules should act as a HBD towards solvents. Work by Kamlet and Taft[17,18] and more recently by Marcus[22,23] have produced values of α and β for a number of liquid solvents. As can be seen in Table 4.2, hydrogen bond donor/acceptor properties are both exhibited by some solvents, whilst some exhibit one, the other or neither.

***Table 4.2**: α and β values for some selected organic solvents at room temperature and 1 bar pressure.*[15-20,22,23]

| **Solvent** | **α** | **β** |
|---|---|---|
| Cyclohexane | 0.00 | 0.00 |
| Triethylamine | 0.00 | 0.71 |
| Carbon Tetrachloride | 0.00 | 0.00 |
| Ethanol | 0.83 | 0.77 |
| Ethyl Acetate | 0.00 | 0.45 |
| Tetrahydrofuran | 0.00 | 0.55 |
| Benzene | 0.00 | 0.10 |
| Methanol | 0.93 | 0.62 |
| Acetic Acid | 1.12 | N/A |
| Acetone | 0.08 | 0.48 |
| Acetonitrile | 0.19 | 0.31 |
| Dichloromethane | 0.30 | -0.014 |
| Pyridine | 0.00 | 0.64 |
| Dimethylformamide | 0.00 | 0.69 |
| Dimethyl Sulfoxide | 0.00 | 0.76 |
| Water | 1.17 | 0.18 |

## 4.1.4 $E_T$ (30) Betaine Dye Scale

The $E_T$ (30) scale is perhaps the most commonly used single parameter indicator series for solvatochromatic studies. The pyridinium *N*-phenolate betaine dye 2,6-diphenyl-4-(2,4,6-triphenylpyridio)-phenolate developed by Reichardt[24,25] is the probe molecule used in this scale. This dye is often referred to as $E_T$ (30) because it was dye number 30 in a series of pyridinium *N*-phenolate betaine dyes tested by Reichardt.[24]




(1)

***Figure 4.2*** *Electronic ground state structure of 2,6-diphenyl-4-(2,4,6-triphenylpyridio)-phenolate,* $E_T$ (30)

The 2,6-diphenyl-4-(2,4,6-triphenylpyridio)-phenolate molecule is largely polarisable due to its extensive aromatic 44 π-electron system. It consists of a strongly hydrogen-bonding/electron pair accepting group in the phenolate oxygen and a weak electron pair acceptor in the nitrogen of the pyridinium group.[25,26] The electronic ground state of the molecule has been measured at $\mu \approx 15$ Debye.[27] This large dipole moment is suitable for the registration of dipole/dipole and dipole/induced dipole interactions

Reichardt's dye exhibit a very high degree of hypsochromacity. This negative

solvatochromism is a result of the differences in the solvation on the ground state and first Franck-Condon excited state. As solvent polarity increases, the difference in the stability of the highly dipolar ground state, and the less dipolar Franck-Condon excited state increases. As can be seen in Figure 4.3, the Franck-Condon excited state is better stabilised in less/non-polar media than in polar media because the dipolarity of the excited state is much less than in the electronic ground state. As this excited state becomes less polar, (~15 Debye[27] to ~6 Debye[28]) non-polar solvents exhibit a greater interaction with the solute hence offering a greater stability than the polar solvents.

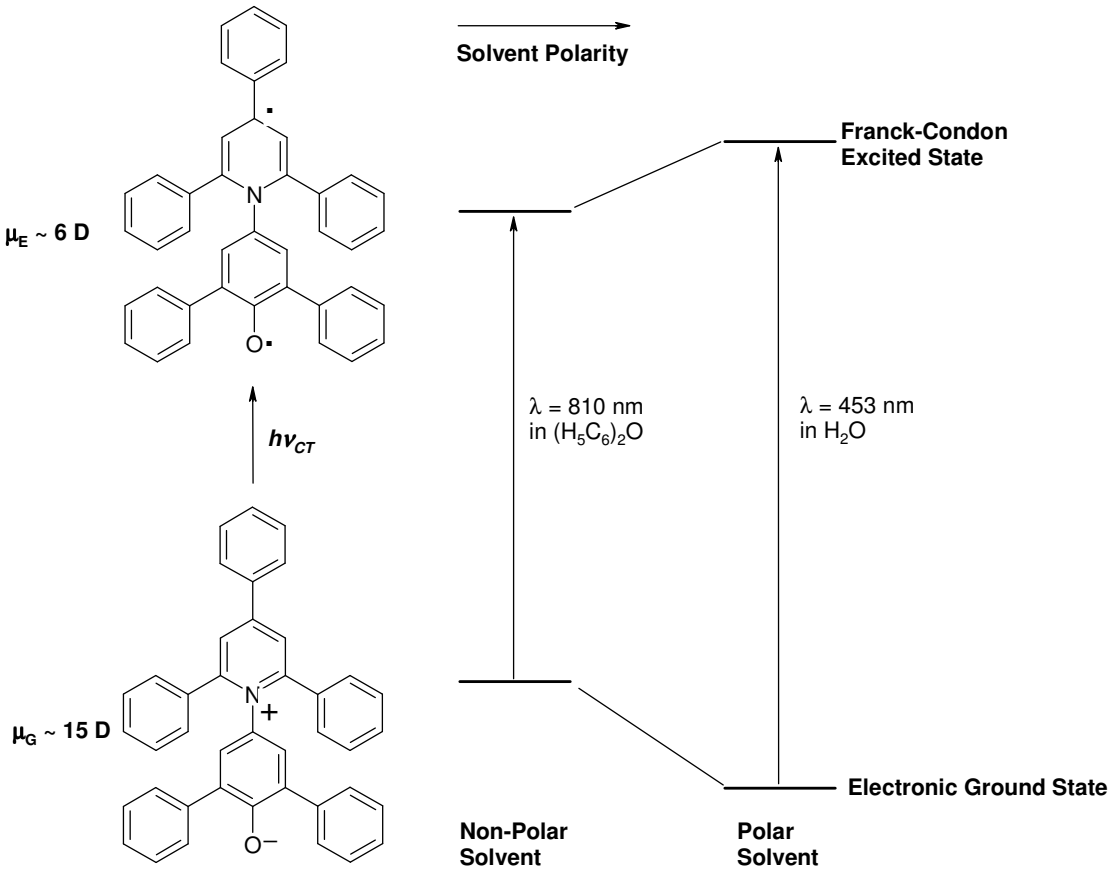



***Figure 4.3*** *Diagrammatic representation of the solvent influence on the intramolecular charge-transfer visible absorption of 2,6-diphenyl-4-(2,4,6-triphenylpyridio)-phenolate ($\mu_G > \mu_E$)*[26]

The transition of $E_T(30)$ dye from the ground state to the Franck-Condon state has been shown to exhibit a change in the dipole moment of ~ 21 Debyes. This is because the Franck-Condon excited-state dipole moment of the $E_T(30)$ dye is antiparallel to that of the ground state, which means that when the ground state is excited by a photon, the

dipole flip surmounts to $\Delta\mu \approx 15$ D - (-6) D = 21 D.[29,30]

The $E_T(30)$ scale is measured analogously to Kosower's $Z$ values.[1,14,24-26]

$$E_T(30)\ (\text{kcal mol}^{-1}) = hc\upsilon_{max}N_A = (2.8591 \times 10^{-3})\ \upsilon_{max}(\text{cm}^{-1}) = 28591/\lambda_{\max} \qquad \textbf{(4.7)}$$

Where $E_T$ is the molar electronic transition energies of dissolved 2,6-diphenyl-4-(2,4,6-triphenylpyridio)-phenolate at 298 K and 1 bar pressure, $\upsilon_{max}$ is the frequency and $\lambda_{max}$ is the wavelength of the longest wavelength with the greatest maximum, $h$ is Planck's constant, $c$ is the speed of light and $N_A$ is Avagadro's number.

In addition to the $E_T(30)$ scale, there is a normalised scale, the $E_T^N$ scale. The $E_T^N$ scale uses water and tetramethylsilane as extreme polar and non-polar reference solvents respectively. In this scale water is given the value 1.00 and tetramethylsilane is given the value 0.00.[24-26] The normalised value for a solvent is calculated by the formula;

$$E_T^N = \frac{E_T(solvent) - E_T(TMS)}{E_T(water) - E_T(TMS)} = \frac{E_T(solvent) - 30.7}{63.1 - 30.7} \qquad \textbf{(4.8)}$$

A list of $E_T(30)$ and $E_T^N$ for some selected molecular liquids are shown in Table 4.3.

***Table 4.3***: *$E_T(30)$ and $E_T^N$ values for some selected organic solvents at room temperature and 1 bar pressure*[1]

| **Solvent** | $E_T$ **(30) (kcal mol$^{-1}$)** | $E_T^N$ |
|---|---|---|
| Tetramethylsilane | 30.7 | 0.000 |
| Cyclohexane | 30.9 | 0.006 |
| Benzene | 34.3 | 0.111 |
| Tetrahydrofuran | 37.4 | 0.207 |
| Pyridine | 40.5 | 0.302 |
| Dichloromethane | 40.7 | 0.309 |
| Acetone | 42.2 | 0.355 |
| Dimethylformamide | 43.2 | 0.386 |
| Acetonitrile | 45.6 | 0.460 |
| Acetic Acid | 51.7 | 0.648 |
| Ethanol | 51.9 | 0.654 |
| Methanol | 55.4 | 0.726 |
| Water | 63.1 | 1.000 |

As this chapter deals with the solvatochromism of Deep Eutectic Solvents, it would be wise to compare results of liquid organic salts and ionic liquids.

### 4.1.5 Polarity of Ionic Liquids

Solvent polarity is defined as being the sum of all possible intermolecular interactions that occur between the solute and the solvent molecules. It therefore should be expected that the polarities of molecular solvents and ionic liquids will differ due to the different degrees of intermolecular interactions experienced arising from the structural differences.

Work carried out using $E_T^N$ scale shows that the addition of altering the size of the alkyl chain on the tetraalkylammonium cation in tetraalkylammonium sulfonate ionic liquids offers a slight increase in the $E_T^N$ value.[30] In comparison, the alkyl chain length of the cation in 1-alkyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide ionic

liquids has virtually no influence on the $E_T^N$ values. The biggest influence on the polarity of the cation in these ionic liquids is the addition of a functional group at the C2 position. Addition of a hydroxyl group significantly increases the $E_T^N$ value, whereas the introduction of a methyl group at said position lowers the value.

Values for π* in ionic liquids is generally high and usually greater than 0.9.[31] The π* values are affected by both the cation and anion. The π* values tend to be higher in ionic liquids than most organic solvents due to the degree of delocalisation of the charge between the two anions. As more atoms are introduced into the anion the charge delocalised becomes more spread out and this results in the value of π* decreasing, due to the decrease in the strength of the Coulombic interactions between the dye indicator molecule and the ion. In the cations, it appears that the decrease in Coulombic interactions between the dye indicator molecule and the ion, caused by delocalising the charge around the imidazolium ring is more than compensated for by the increased polarizability of the delocalised system.[31] This observation is backed up by the relatively high π* value seen in toluene (*c.f.* 0.532). Both the anion and the cation show that they both contribute to the polarisability parameter value. The α parameter is mainly governed by the cationic component of the ionic liquid. For example, values for $[BMIM]^+$ based ionic liquids have moderately high α values, > 0.6, similar to *t*-BuOH, with very small changes being observed when the anion is changed.[20,31] The β parameter is more dependant on the anion. Ionic liquids in general tend to show higher β values in comparison to other solvents than might be expected since the anions are effectively conjugate bases of strong acids, which would suggest lower β values should be observed, however this is not the case.

By using the solvatochromic probe Nile red the polarities of various [BMIM] ionic liquids have been shown to be related to the degree of charge. By changing the anion of [BMIM], Carmichael and Seddon have shown the polarity decreases in the order of $[NO_3]^- > [BF_4]^- > [(CF_3SO_2)_2N]^- > [PF_6]^-$.[33] On first inspection this trend does not appear to agree with a decrease in polarity as the anion size increases since $[(CF_3SO_2)_2N]^-$ is obviously the larger anion. However, there is crystallographic evidence from an analogous imidazolium $[(CF_3SO_2)_2N]^-$ salt that reveals the negative charge is only partially delocalized within the anion. The results show that the negative charge is only delocalised over the S—N—S moiety, which is larger than the tetrafluoroborate anion but

smaller than the hexafluorophosphate anion. Anionic effects on solvent polarity in ionic liquids become negligible when cation size increases. In the case of [HMIM] and [OMIM] ionic liquids, it was reported that there was s no difference between the $[BF_4]$ and $[PF_6]$ series.[33] It has been shown that as the length of the 1-alky group on the imidazolium cation is increased the polarity increases. A maximum is reached at $C_6$, followed by a decrease in polarity as the length of the chain continues to increase. Table 4.4 shows selected ionic liquids and their respective $E_T^N$ ,α, β and π* values.

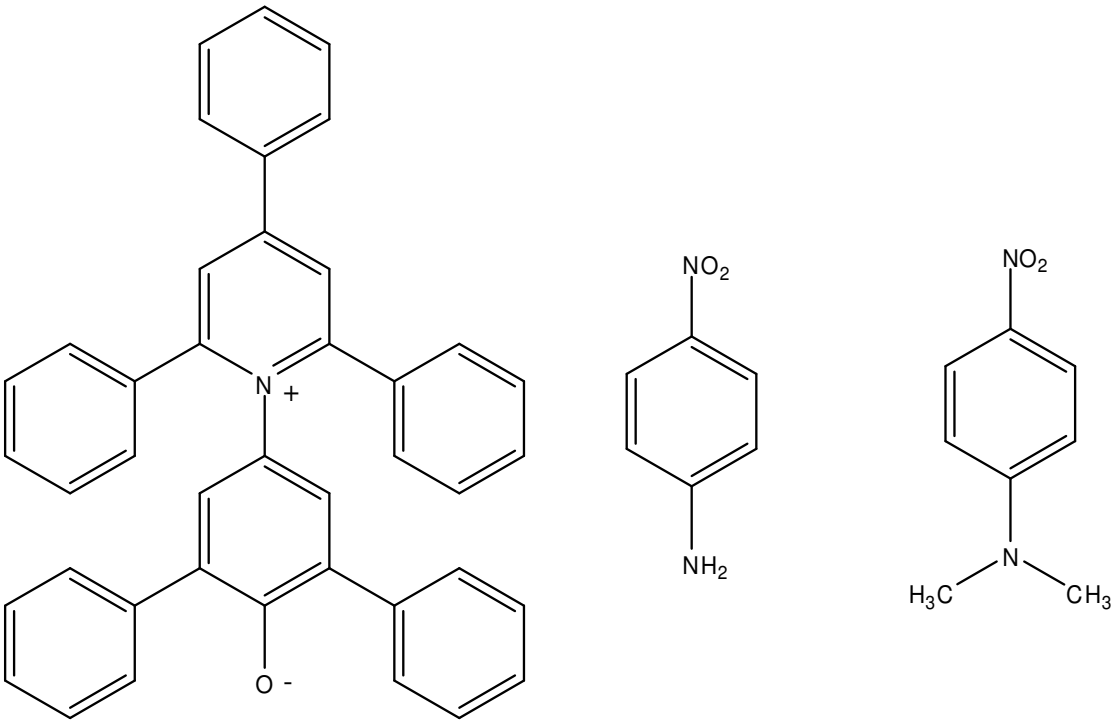
***Table 4.4**: $E_T^N$ , α, β and π* values for some selected ionic liquids at room temperature and 1 bar pressure in ascending order*

| Ionic Liquid | $E_T^N$ | α | β | π* |
|---|---|---|---|---|
| $[ethylammonium]^+ [NO_3]^-$ | 0.95 | 0.85 | 0.46 | 1.24 |
| $[n\text{-octyl-pyridinium}]^+ [Tf_2N]^-$ | 0.588 | 0.51 | 0.28 | 0.99 |
| $[2\text{-methyl octylpyridinium}]^+ [Tf_2N]^-$ | 0.554 | 0.48 | 0.35 | 0.95 |
| $[3\text{-methyl octylpyridinium}]^+ [Tf_2N]^-$ | 0.576 | 0.50 | 0.33 | 0.97 |
| $[1\text{-butyl-2,3-dimethylimidazolium}]^+ [BF_4]^-$ | 0.541 | 0.381 | 0.239 | 1.010 |
| $[1\text{-butyl-3-methylimidazolium}]^+ [BF_4]^-$ | 0.630 | 0.627 | 0.376 | 1.047 |
| $[1\text{-butyl-3-methylimidazolium}]^+ [PF_6]^-$ | 0.669 | 0.634 | 0.207 | 1.032 |
| $[1\text{-butyl-3-methylimidazolium}]^+ [TfO]^-$ | 0.656 | 0.625 | 0.464 | 1.006 |
| $[1\text{-butyl-3-methylimidazolium}]^+ [N(Tf)_2]^-$ | 0.644 | 0.617 | 0.243 | 0.984 |
| $[n\text{-butylmethyl pyridinium}]^+ [N(Tf)_2]^-$ | 0.544 | 0.427 | 0.252 | 0.954 |

## 4.2 Results and Discussion

To calculate the π*, α and β values three indicator probe molecules were used. The molecules used to investigate these parameters were Reichardt's Dye 30 (1), 4-nitroaniline (2) and N,N-dimethyl-4-nitroaniline (3). For this work, the solvents investigated will be standardized around these single probe sets. This is common practice with many workers in the field. Other workers in this field however favour the use of

several probes and then average the values obtained. The disadvantage of using many probes is that considerable effort is required to derive parameters for any new solvent. The single probe set approach has the advantage of being experimentally less involved, but also the data obtained will be comparable with published literature.[1]




(1) (2) (3)

***Figure 4.4*** *Diagrammatic representation of dyes used in calculating π*,α and β*

The main purpose of this chapter is to understand the solvent properties of several different alcohol based Deep Eutectic Solvents used throughout this study and to determine whether polarity parameters can explain the product distributions observed for a series of esterification reactions. Several different parameters will also be investigated such as choline chloride concentration and position of the alcohol groups along the carbon chain. The intrinsic error of the instrumentation was calculated to be approximately +/- 2%.

### 4.2.1 $E_T$(30) Betaine Dye Scale Results

The values for $E_T$(30) for several Deep Eutectic Solvents used in this project can be seen in Table 4.6. As reported in section 4.1.4 the $E_T$(30) value was calculated from the $\lambda_{max}$ value using the formula $E_T$(30) (kcal mol$^{-1}$) = 28591/$\lambda_{max}$, with $E_T^N = \frac{E_T(solvent) - 30.7}{32.4}$

***Table 4.5***: *$E_T$(30) data obtained for several Deep Eutectic systems ( $\Delta E_T$(30) values compare the pure alcohol with that containing ChCl)*

| **Solvent** | $E_T$ **(30) /kcal mol$^{-1}$** | $E_T^N$ | $\Delta E_T$ **(30) /kcal mol$^{-1}$** |
|---|---|---|---|
| Glycerol | 57.17 | 0.817 | N/A |
| ChCl:Glycerol 1:3 | 57.96 | 0.841 | 0.79 |
| ChCl:Glycerol 1:2 | 58.58 | 0.860 | 1.41 |
| ChCl:Glycerol 1:1.5 | 58.21 | 0.849 | 1.04 |
| ChCl:Glycerol 1:1 | 58.49 | 0.858 | 1.32 |
| Ethylene Glycol | 56.38 | 0.793 | N/A |
| ChCl:Ethylene Glycol 1:2 | 57.75 | 0.835 | 1.37 |
| 1,4-Butanediol | 53.53 | 0.705 | N/A |
| ChCl:1,4-Butanediol | 54.76 | 0.743 | 1.23 |
| 1,3-Butanediol | 52.62 | 0.677 | N/A |
| ChCl:1,3-Butanediol | 56.72 | 0.803 | 4.1 |
| 1,2-Butanediol | 52.93 | 0.686 | N/A |
| ChCl:1,2-Butanediol | 55.41 | 0.763 | 2.48 |
| ChCl: Urea 1:2 | 58.34 | 0.853 | N/A |

The data in Table 4.5 shows that the choline chloride based DES's exhibit a polarity that is similar to $RNH_3^+X^-$ and $R_2NH_2^+X^-$ ionic liquids with discrete anions.[26] The diols and triol have $E_T$(30) values that are similar to those for other alcohols although the values are in general slightly higher (*c.f.* Table 4.3) It can also be seen that the addition of choline chloride increases the $E_T$(30) of the system. The magnitude of the change in

$\Delta E_T(30)$ is unusual. The maximum change observed in $\Delta E_T(30)$ is that for the addition of ChCl to 1,3-butanediol which results in a change of approximately 4.1 kcal $mol^{-1}$. This is only comparable to the difference between pure glycerol and pure 1,3-butanediol. While this may seem anomalous it must be appreciated that in the eutectic mixture a proportion of the alcohol is replaced by the salt ions which in themselves are larger than the alcohol moieties.

This can be observed qualitatively by comparing the surface potentials of $Ch^+$ and glycerol as shown in Figure 4.5.

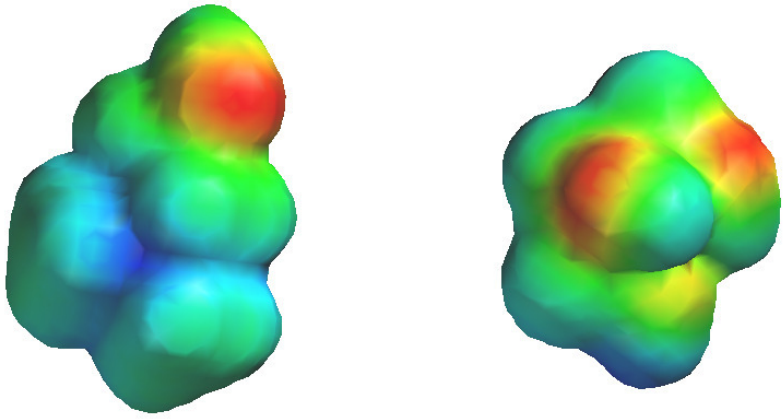


***Figure 4.5**: Surface potentials of $Ch^+$ and glycerol respectively, calculated using commercial software.*[34]

It could be argued that polarity in these systems is governed by the density of OH groups. Using the data calculated for the molar volume in the previous chapter it is possible to calculate a molar volume per OH group. This should correlate with the polarity parameters if dominated by hydrogen bond formation. Figure 4.6 shows that the $E_T(30)$ values for the pure liquids correlate well with this parameter. The eutectic mixtures also show a roughly linear trend, but it is not co-linear with the pure alcohols. This could suggest that the difference is due to the presence of the chloride ions which cause an approximate 3 kcal $mol^{-1}$ increase in $E_T(30)$.

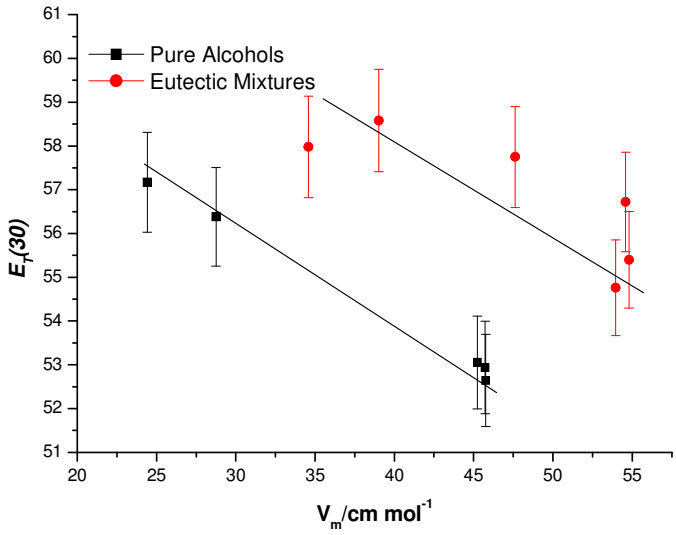



***Figure 4.6:*** *Correlation between $E_T(30)$ and the molar volume per OH group*

## 4.2.2 Polarisability and Hydrogen Bond Acceptor/Donor Parameters π*, α and β

Various methods have been used to calculate the polarity parameters from the Kamlet and Taft equation described earlier.[15-20] However, each method gives slightly different values. A multi-parameter fit depends significantly on the solvents used to calibrate the dyes and the literature values used in the fit. The main aim of this work is to be able to compare eutectic based ionic liquids with those containing discrete anions. Consequently, it was decided that it was most appropriate to follow the literature method used in ionic liquid systems such that the parameters would be comparable.[35-39] To calculate π* the following formula was used with the data obtained from the indicator molecule 4-nitroaniline

$$\pi^* = (\nu_{solvent} - \nu_{cyclohexane})/(\nu_{dmso} - \nu_{cyclohexane}) \quad \textbf{(4.9)}$$

α values were calculated using $E_T(30)$ data and 4-nitroaniline π* data using equation (4.5).

$$\alpha = (E_T(30) - 14.6(\pi^* - 0.23) - 30.31)/16.5 \quad \textbf{(4.10)}$$

The β values were calculated from the data obtained for 4-nitroaniline and N,N-dimethyl-4-nitroaniline

$$\beta = 0.9\,(\Delta\nu_{cyclohexane} - \Delta\nu_{solvent})/(\Delta\nu_{cyclohexane} - \Delta\nu_{propan\text{-}1\text{-}ol}) \quad \textbf{(4.11)}$$

The resultant parameters are listed in Table 4.6 and the raw data is listed in Table 4.7.

***Table 4.6:*** *π*,α and β data for several Deep Eutectic Solvents*

| Solvent | π* | α | β |
|---|---|---|---|
| Glycerol | 0.956 | 0.882 | 0.658 |
| ChCl:Glycerol 1:3 | 0.970 | 0.914 | 0.657 |
| ChCl:Glycerol 1:2 | 0.984 | 0.937 | 0.657 |
| ChCl:Glycerol 1:1.5 | 0.980 | 0.921 | 0.658 |
| ChCl:Glycerol 1:1 | 1.003 | 0.923 | 0.658 |
| Ethylene Glycol | 0.890 | 0.877 | 0.696 |
| ChCl:Ethylene Glycol 1:2 | 0.970 | 0.903 | 0.687 |
| 1,4-Butanediol | 0.915 | 0.722 | 0.764 |
| ChCl:1,4-Butanediol | 1.174 | 0.646 | 0.791 |
| 1,3-Butanediol | 0.856 | 0.707 | 0.796 |
| ChCl:1,3-Butanediol | 0.977 | 0.848 | 0.750 |
| 1,2-Butanediol | 0.846 | 0.728 | 0.828 |
| ChCl:1,2-Butanediol | 0.941 | 0.802 | 0.748 |
| Reline | 0.990 | 0.922 | 0.642 |

***Table 4.7:*** *$\lambda_{max}$ data for several Deep Eutectic Solvents*

| Solvent | 4-nitroaniline $\lambda_{max}$ /nm | N,N-dimethyl-4-nitroaniline $\lambda_{max}$ /nm | $E_T(30)$ $\lambda_{max}$ /nm |
|---|---|---|---|
| Glycerol | 386.3 | 413.1 | 500.0 |
| ChCl:glycerol 1:3 | 387.2 | 414.1 | 493.2 |
| ChCl:glycerol 1:2 | 388.0 | 414.9 | 489.0 |
| ChCl:glycerol 1:1.5 | 387.8 | 414.6 | 491.1 |
| ChCl:glycerol 1:1 | 389.2 | 416.0 | 488.7 |
| 1,2-ethanediol | 382.3 | 406.2 | 507.0 |
| ChCl:1,2-ethanediol 1:2 | 387.2 | 411.8 | 494.9 |
| 1,4-butanediol | 383.8 | 402.4 | 534.3 |
| ChCl:1,4-butanediol 1:2 | 399.6 | 416.1 | 522.5 |
| 1,3-butanediol | 380.2 | 396.3 | 540.1 |
| ChCl:1,3-butanediol 1:2 | 387.6 | 407.3 | 515.9 |
| 1,2-Butanediol | 379.6 | 393.2 | 543.2 |
| ChCl:1,2-Butanediol 1:2 | 385.4 | 405.2 | 504.0 |
| ChCl:urea 1:2 | 388.4 | 416.5 | 490.2 |

As with the $E_T(30)$ values described in section 4.2.1 the magnitude of the π*, α and β values are very similar to those reported for the imidazolium based ionic liquids. Again it is surprising how little the values vary firstly from one another and secondly when compared to the pure alcohols.

In an analogous manner to that described for $E_T(30)$ above, a correlation should exist between the molar volume per OH group and some of the polarity parameters. Figure 4.7 shows the correlation for α with the molar volume per OH group and it is evident that the same correlation as seen in Figure 4.6 is present. This is slightly unexpected, however, since it could be argued that the addition of ChCl would cause a decrease in H-bond donor properties since the $Cl^-$ would act as a strong H-bond acceptor. One possible explanation for this could be that the $Cl^-$ acts as a structure breaking species as discussed in Chapter 3. This could disrupt the strong inter-glycerol interactions and permit more interaction with the indicator solute. There is also a similar correlation for the β parameter which can be seen in Figure 4.8.

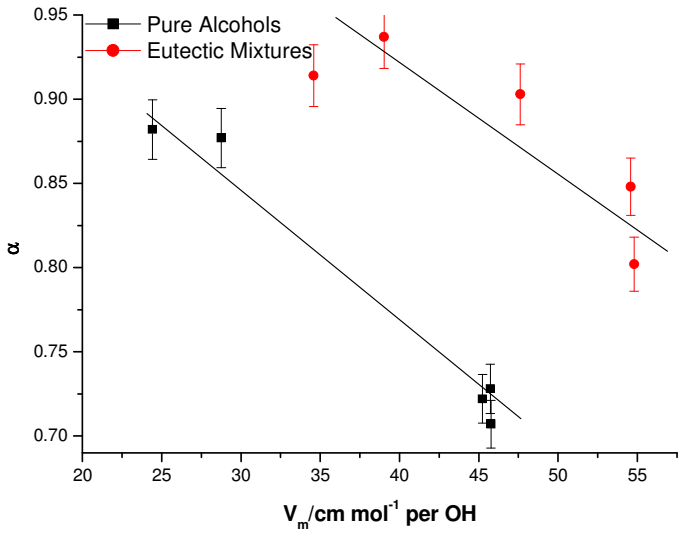



***Figure 4.7:*** *Correlation between α and the molar volume per OH group*

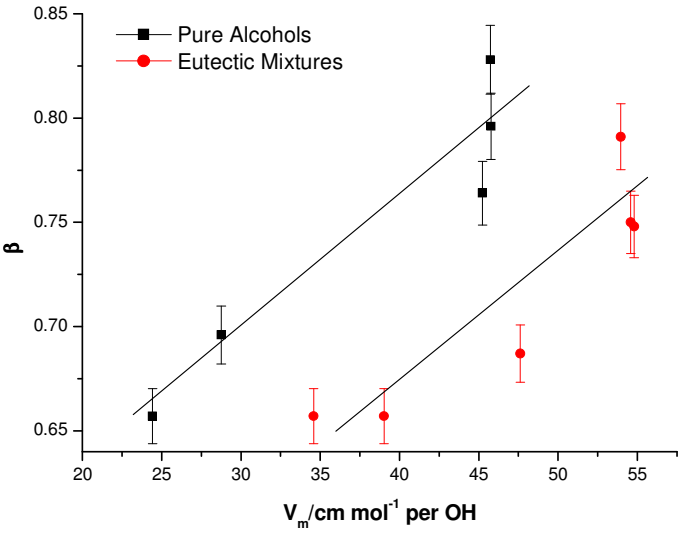



***Figure 4.8:*** *Correlation between β and the molar volume per OH group*

However, there is negligibe correlation between π* and the molar volume per OH group showing that this simple model for the solvation in these eutectic systems is far from accurate.

## 4.2.3 Effect of Choline Chloride Concentration

To measure the effect of choline chloride concentration on the polarity of a system, the system most stable to containing concentrations of choline chloride above the eutectic point needed to be used. In this case glycerol based systems were used. Although it has been shown in Chapter 3 the eutectic point of ChCl and glycerol is in a 1:2 ratio, higher levels of ChCl can be added without phase separation at room temperature. **Figure 4.8** shows a roughly linear increase in $E_T(30)$ with ChCl concentration. Extrapolating this trend to 100% ChCl gives a value of approximately 59 kcal mol$^{-1}$ which is comparable to that for ethylammonium nitrate (Table 4.4) which would seem reasonable.

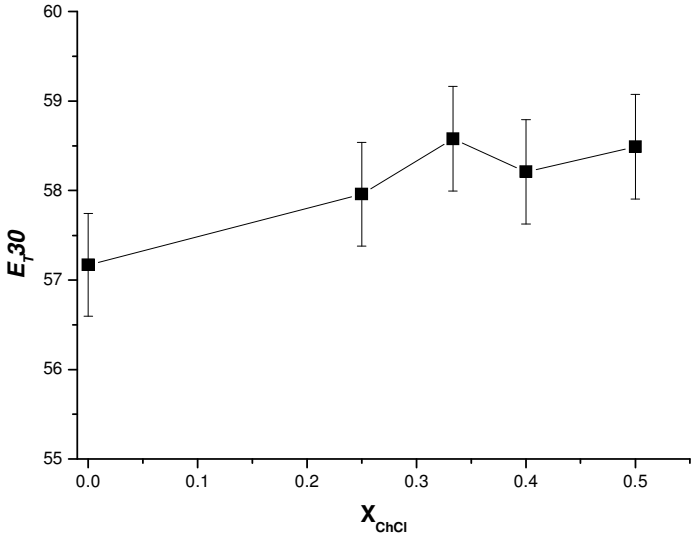



***Figure 4.9:*** *$E_T(30)$ as a function of choline chloride concentration for the glycerol base systems*

Similar tends are observed for π*, α and β although the parameters are relatively insensitive to changes in polarity. The β parameter in particular does not appear to show significant changes with increasing chloride concentration. To some extent this is not surprising since the number glycerol OH groups is present in excess and will act as a better hydrogen bond donor than the indicator solute. It is proposed that in high polarity solvents solvatochromic indicators are insensitive to compositional changes because solvent-solvent interactions dominate over solvent-solute interactions.

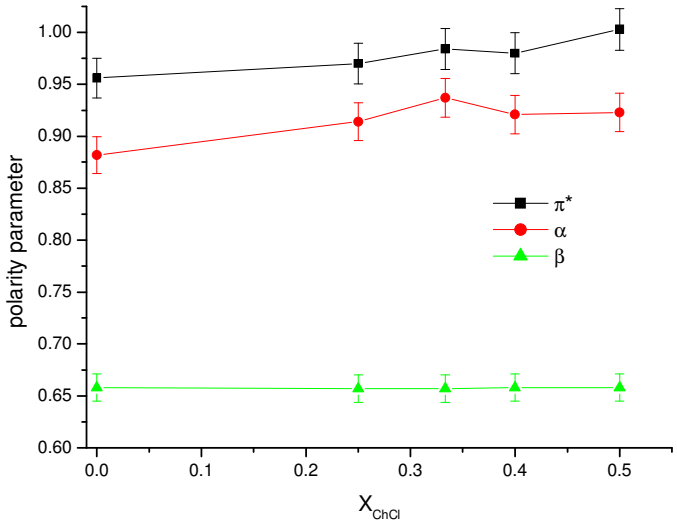



***Figure 4.10:*** *Variation of π*, α and β with mole fraction of ChCl in glycerol*

***Table 4.7:*** *Change in solvatochromic parameters as a function of ChCl concentration*

| System | Δπ* | Δα | Δβ | Δ $E_T$ (30) |
|---|---|---|---|---|
| **ChCl:Glycerol 1:3** | 0.015 | 0.032 | -0.001 | 0.79 |
| **ChCl:Glycerol 1:2** | 0.028 | 0.056 | -0.001 | 1.41 |
| **ChCl:Glycerol 1:1.5** | 0.025 | 0.039 | 0.000 | 1.04 |
| **ChCl:Glycerol 1:1** | 0.048 | 0.041 | 0.000 | 1.32 |
| **ChCl:Ethylene glycol 1:2** | 0.080 | 0.026 | -0.009 | 1.37 |
| **ChCl:1,4-Butanediol** | 0.259 | -0.076 | 0.027 | 1.23 |
| **ChCl:1,3-Butanediol** | 0.121 | 0.074 | -0.046 | 4.10 |
| **ChCl:1,2-Butanediol** | 0.095 | 0.140 | -0.080 | 2.48 |

## 4.3 Conclusions

This work has shown that the interactions between the glycols and ChCl are complex. The polarity parameters of the mixtures are dominated by the properties of the glycols and there is little effect of ChCl on the hydrogen bond donor or acceptor properties of the glycerol based liquids. There is however a more drastic effect of ChCl